

POLICY ARTICLE

REGULATORY IMPACT

Correcting the logic behind the 2026 US vehicle emissions standard rollback

Correction of selective assumptions and inconsistent interpretations of evidence would overturn policy justification

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Vehicle emission and fuel economy standards—the sector’s most important policies—have swung wildly in the US since 2000. Modest tightening of standards under the Bush administration was accelerated under Obama, partially rolled back under the first Trump administration, then reinstated and tightened under Biden. Yet the current Trump administration has gone further, with a complete elimination of the US Environmental Protection Agency (EPA) greenhouse gas (GHG) standards for vehicles. These regulatory swings each claim support from formal cost-benefit analysis, as required by executive orders. Yet the contradictory conclusions stem not from evolving evidence, but from selective assumptions and inconsistent interpretation of key evidence. We discuss how to interpret and apply the existing evidence correctly and demonstrate that correcting errors would overturn the justification for the elimination of the EPA GHG standards.

Light-duty vehicles account for over half of US transportation GHG emissions and 16% of total emissions (*1*). EPA regulations, our focus, regulate GHG emissions, but the US Department of Transportation (DOT) separately regulates fuel economy. The current administration zeroed out penalties for the DOT regulation and has proposed a weak version that is unlikely to be binding on industry, which de facto leaves the market with no effective regulation. The recent rollback of EPA standards under the second Trump administration took effect in April 2026 and will surely be disputed in court, where errors in analysis could create legal vulnerabilities if they undermine the agency’s stated justification for the rescission.

There are two crucial changes in the recent cost-benefit analysis. One is the decision by EPA to ignore environmental benefits of the GHG standards, eliminating what was \$1.8 trillion in benefits from reducing GHGs and copollutants according to the analysis of the 2024 standard. This change is transparent and has been extensively critiqued. But there is a second key change that is on the same order of magnitude as the first but is less transparent and more difficult to assess. This change relates to the treatment of how consumers value fuel savings and vehicle attributes. Both the elimination of environmental benefits and the change in valuation are alone enough to reverse whether the rollback has positive net benefits. We focus on the valuation change because of its large magnitude and the fact it has received relatively little attention, but we recognize the importance of both.

The core of the valuation issue is that research consistently shows that new car buyers are willing to pay less than \$1 extra for the car in exchange for \$1 in discounted fuel savings (2–8) [see supplementary materials (SM) A]. This finding admits multiple interpretations,

such as reflecting behavioral biases that imply the low valuation is a mistake, or hidden costs that occur if vehicle efficiency is improved at the expense of other vehicle attributes. The cost-benefit analysis hinges on the relative importance of these different mechanisms. In the case of hidden costs from worsened attributes, we show how the losses to consumers will also be mirrored in technology cost savings for producers. Past administrations have toggled between extremes on how to treat the issues around valuation, sometimes without clear explanation. When an agency makes a strategic choice on this issue, it can swing analysis substantially. We show how the most recent change strips many billions of dollars in benefits from regulating vehicle GHG emissions.

We provide a framework that maps empirical research on consumer valuation into cost-benefit analysis consistently and transparently (see SM B). We apply this framework to the EPA’s 2026 Regulatory Impact Analysis (RIA), which puts emphasis on a scenario that takes an extreme position and credits only \$0.22 out of every \$1 of fuel savings as a benefit (*9*). We show that both this position and the approach taken by EPA under the Biden administration are premised on a misreading of the evidence.

Consumer valuation deserves focus for several reasons. It is pivotal: Proper treatment would overturn the current administration’s analysis. By contrast, it would have only modestly affected the Biden-era 2024 analysis. It is also mishandled: Multiple recent administrations have made modeling errors or misapplied the available empirical evidence, enabling policy swings that paralyze long-term industry investment. Finally, it is opaque: Unlike transparent parameters like gasoline prices or cost per ton of carbon emissions, the treatment of fuel savings valuation remains buried in analytical confusion. Our framework brings needed clarity. The framework is also relevant to energy efficiency standards promulgated by the Department of Energy, where the same valuation issues arise.

A FRAMEWORK FOR HOW CONSUMERS VALUE FUEL SAVINGS AND VEHICLE ATTRIBUTES

The starting point to understand valuation is the question: How much more are consumers willing to pay, relative to a benchmark vehicle, for a vehicle that saves \$1 in fuel costs (applying a discount rate to express future costs in present dollars)? Consumers may be willing to pay less than \$1 for two reasons. First, consumers may systematically undervalue fuel savings owing to inattention, misperceptions about future fuel use or price, a bias in favor of the present relative to the future, other cognitive biases, or credit constraints. Second, there may be attribute trade-offs: To reduce vehicle cost, manufactur-

ers may choose designs that save fuel but compromise other attributes (weight, performance, comfort, etc.). Our framework (see SM B) shows how every dollar of fuel savings can be decomposed into three components that must sum to one: (i) willingness to pay for the fuel-efficient vehicle; (ii) inattention or other behavioral biases leading to undervaluation of fuel savings; and (iii) attribute trade-offs reducing consumer utility (“hidden costs”).

In one edge case, suppose that automakers respond to regulation by deploying a fuel-saving technology that changes no other attribute of a car, and that consumers are fully attentive to fuel costs and have no biases. In that case, consumers will value \$1 of savings at \$1. In reality, many fuel-efficiency improvements affect other vehicle attributes, negatively or positively depending on the specific technology (10). Economic evidence also shows that many buyers imperfectly value future savings (2–8). Thus, in reality we expect consumers to be willing to pay less than \$1 per \$1 in fuel savings and that the gap will be a mix of inattention (or other biases) and hidden costs. But logically, the components sum to one. Past regulatory analyses have made inconsistent, and often implicit, assumptions about these components. Using an explicit decomposition—with each component justified by credible evidence and constrained to sum to one—would greatly improve analytical transparency.

VALUATION IN THE 2026 EPA ANALYSIS

The EPA’s 2026 RIA concludes that eliminating vehicle greenhouse-gas standards would increase US welfare by \$600 to \$790 billion (9). This conclusion rests on a problematic approach to valuation and the associated technology costs. We map both the earlier 2024 EPA analysis, which supported stricter standards, and the 2026 analysis, into our valuation framework (10, 11). We reveal implicit assumptions of both analyses that run counter to empirical evidence.

Both recent EPA analyses assume car buyers require a 2.5-year payback period, and thus include only the first 2.5 years of fuel savings in benefit calculations. This implies a willingness to pay for \$1 in fuel savings of \$0.21 to \$0.25 (see SM A.1). The 2026 RIA introduces a scenario with no undervaluation of future fuel savings. Consequently the remainder—\$0.75 to \$0.79—is assumed by EPA to represent the hidden attribute trade-offs. These assumptions contradict the preponderance of available evidence, as discussed below. Likewise, although industry convention cites similar 2.5-year payback periods, automakers typically associate the remainder with undervaluation rather than attribute trade-offs—contrary to the 2026 RIA’s assumption. By contrast, the 2024 analysis assumes no hidden attribute trade-offs, assigning the full remainder to inattention.

The assumptions about valuation in the present cost-benefit analysis also create an internal logical inconsistency: Both the 2024 and the 2026 RIA estimate technology costs using established methodologies that assume fuel-saving technologies hold other attributes (weight, performance, comfort, etc.) constant. Yet the 2026 RIA simultaneously assumes large attribute trade-offs that produce hidden costs. If automakers achieved fuel economy by sacrificing performance or weight rather than deploying new technology, costs would be substantially lower than the 2026 RIA estimates (SM A.3).

AVAILABLE EMPIRICAL ESTIMATES

The assumption that consumers value only 2.5 years of savings lacks empirical grounding (see SM A.1). The academic literature estimates either undervaluation or attribute trade-offs and uses these to infer willingness to pay. We know of no credible studies directly estimating willingness to pay in regulatory contexts involving trade-offs. Nevertheless, absent direct estimates, we accept the RIA’s implied willingness to pay of \$0.21 to \$0.25 as our starting point.

The 2026 RIA’s assumption of no undervaluation of future fuel savings is inconsistent with a literature review of recent high-quality studies finding substantial undervaluation (2–8). The analysis fur-

Monetized impacts of rescinding standards

Values are calculated in billions of 2024 dollars. Columns 2 to 4 reflect adjustments to the US Environmental Protection Agency (EPA) values in column 1. The combined change in net benefits after all adjustments equals \$1.46 trillion at a 3% discount rate. Pollution benefits of the standards are not included.

	2026 EPA scenario (A3) (1)	Adjustment: rebound (2)	Adjustment: fuel-savings split (3)	Adjustment: technology costs (4)
3% DISCOUNT RATE				
Savings	1290	1290	1290	748
Costs	500	734	1418	1418
Net savings	790	556	-128	-670
7% DISCOUNT RATE				
Savings	850	850	850	493
Costs	240	365	732	732
Net savings	600	485	118	-239

ther distorts results by applying high discount rates to future payoffs and ignoring an approximation bias present in some studies. Our literature summary (see SM A.1 and table S2) addresses these issues and indicates inattention (or other behavioral biases) between \$0.54 and \$0.62.

Regarding attribute trade-offs, the RIA emphasizes the gasoline-to-electric transition as an important burden to consumers, yet available empirical evidence suggests they face much smaller costs. A recent study estimates attribute trade-offs of fully electric vehicles (EVs) compared to their conventional gasoline vehicle counterparts at between \$0.17 and \$0.31 per dollar of the EV’s fuel cost savings, declining over time as EV range improves (12). Further, these are upper bounds: Consumers need not choose the fully electric version of the vehicle they would have purchased absent regulations and will only switch to other technologies (e.g., hybrids) or other vehicles if they value them more.

Taken together, our suggested ranges for the three components—willingness to pay, inattention or other biases, and attribute trade-offs—are 21 to 25%, 54 to 62%, and 17 to 31%, respectively. Although determined separately, the midpoints of these ranges sum up to 100% almost exactly. For the current year, we propose that the EPA use \$0.22, \$0.55, and \$0.23 for consumer valuation, inattention or other biases, and attribute trade-offs, respectively. The value for attribute trade-offs should decline over time. Both the 2024 and 2026 estimates attributed \$0.22 to willingness to pay, as does our estimate. However, the 2024 estimate attributed \$0.78 to inattention and \$0 to trade-offs, whereas the 2026 estimate attributed \$0 to inattention and \$0.78 to trade-offs. In stark contrast to those earlier EPA analyses, our estimate attributes \$0.23 to trade-offs and \$0.55 to inattention (fig S1) (10, 11).

FUEL PRICES, DRIVING CHANGES, AND ELECTRIC VEHICLES

We have additional substantive concerns with the regulatory analyses beyond valuation. The 2026 Rule makes asymmetric assumptions about fuel prices, presenting the central fuel price forecasts from the 2025 Annual Energy Outlook alongside a low-price case, without presenting the corresponding high-price scenario—all of this in a current gasoline price environment that well exceeds the central case. Assuming low gasoline prices will mechanically and unrealistically lower the benefit of fuel economy improvements (SM A.2).

The 2026 RIA analysis also makes an accounting error associated with reductions in driving when cars become more expensive to operate (a reverse of what is often called the “rebound effect”) (13). Specifically, the analysis says that consumers are better off when driving

is more expensive because the higher cost leads consumers to drive less, creating large savings in fuel, maintenance, and repair costs from reduced trips, which EPA estimates at around \$250 billion. By basic economic logic, however, consumers valued those lost trips by at least as much as they would have paid to take them, and so the foregone value of those trips must also be included as a cost of rescission (SM A.4). The magnitude of the change in driving itself, which EPA estimates at about 1.4 trillion miles lost over the analysis period, is debated in the literature (13), so we do not examine it here. However, regardless of the estimated change in driving, the cost-benefit analysis must be consistent with the logic of why consumers choose to drive.

We also note that both the 2024 and 2026 analyses ignore potential electricity distribution costs that could be triggered by increased EV penetration. The EV transition could also potentially deliver benefits by smoothing demand on the electric grid. These costs and benefits are prospective, and therefore harder to estimate, but we believe they could be substantial. Analyses of standards that involve EV adoption should be expanded to include any notable system costs associated with vehicle electrification, as well as any gasoline-related infrastructure costs that occur as a result of eliminating the standards.

OVERALL IMPLICATIONS FOR COST-BENEFIT ESTIMATES

Applying the accounting correction for driving changes, our proposed undervaluation estimates, and the associated changes in technology cost leads to a substantially different conclusion than that in the 2026 RIA. Starting from the EPA's full-valuation scenario A3 (see the Table, column 1), we first estimate that correcting for driving changes leads to a \$234 billion increase in the costs of rescinding the EPA GHG standards (see the Table, column 2 versus 1). Next, we apply our proposed division of fuel savings: The costs of rescission increase further by \$684 billion (see the Table, column 3 versus 2). Thus, if we simply apply EPA's own modeling and adjust for these corrections, we find that the conclusion of the 2026 EPA analysis is reversed.

Finally, there is a logical inconsistency in the 2026 RIA of measuring technology costs assuming manufacturers use fuel-saving technologies that hold other attributes constant, while at the same time including large hidden costs of attribute trade-offs in valuation. This inconsistency overstates technology costs in the 2026 RIA; automakers will choose technologies with hidden costs only if they reduce compliance costs. Estimates from the literature imply that this inconsistency could inflate the savings from rescinding the standards by 42% or \$542 billion (see the Table, column 4 versus 3). Added together, we conclude that these three corrections alone reduce the net benefit of rescinding the standards by \$1.5 trillion.

The elimination of environmental benefits is also deeply problematic from an economics perspective, and this is valued at \$1.8 trillion in the 2024 analysis. Taken together, we conclude that a more careful and complete analysis would almost certainly come to a different conclusion than in the 2026 RIA.

ALTERNATIVE POLICIES FOR VEHICLES

Analyses of regulation typically give only perfunctory consideration to policy alternatives. A serious consideration of alternatives, which could identify policies to reduce GHG emissions at lower cost, would be a positive outcome of the current debate and policy whiplash.

New vehicle taxes based on fuel consumption or emissions factors offer one alternative to corporate average fuel economy (CAFE). Whether structured as differentiated taxes or as "feebates" (combining taxes on less efficient vehicles with rebates for more efficient vehicles), these instruments make compliance costs transparent—unlike CAFE, where costs are hidden in design and pricing choices. Opacity may be advantageous for political reasons, but it creates confusion by enabling stakeholders to exaggerate or understate costs. Fuel

consumption-based taxes eliminate this ambiguity, address known limitations of performance standards regarding market size, and can raise revenue if desired. Dozens of countries already implement such systems successfully.

Fuel or mileage taxes provide another alternative with considerable advantages. Unlike CAFE, fuel pricing (as would be the case with a carbon tax) affects both vehicle choice and driving behavior, encourages retirement of older vehicles, and immediately affects the entire fleet when adjusted. CAFE standards, by contrast, take decades to permeate the vehicle stock. The main limitation is that if consumers ignore fuel prices when purchasing vehicles, they may underinvest in efficiency without standards (14). When undervaluation of future fuel savings is substantial, as our review suggests, a hybrid approach combining fuel taxes with standards may be preferable.

The 2026 rollback rests on several fundamental errors. Correcting the issues in how the valuation of future fuel savings is treated would alone negate the economic justification for EPA's decision to rescind the standards. When applying further corrections for other problematic assumptions, most notably the omission of environmental damages from the analysis, the case for removing the standards in the 2026 EPA analysis is even more clearly unjustified. Policy volatility hamstringing automakers' long-term investments. Although political power drives these swings, rigorous and transparent cost-benefit analysis—consistently applied—could moderate them. Application of our framework to deal with valuation could correct errors from multiple administrations, providing analytical consistency that could help improve and stabilize policy. □

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SUPPLEMENTARY MATERIALS

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